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Project Title:	Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys		
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	EDITOR's NOTE (12/14/2015): Continuation of "Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys--NNX14AB34G; Principal Investigator: Alain Karma, PhD." Cellular and dendritic microstructures formed during industrial solidification processes (such as casting, welding, and brazing) determine the properties of commercial metallic alloys produced in numerous industries, with applications ranging from automobile engine parts to aircraft turbine blades. The production of innovative materials and the optimization of both properties and processing routes go through an accurate prediction and modeling of the dynamics of microstructures formation. From a fundamental standpoint, the formation of these structures is controlled by a subtle interplay between microscopic interfacial phenomena and macroscopic transport of heat and solute, which is strongly		

Task Description:	influenced by gravity-induced melt convection under terrestrial conditions. Therefore, microgravity experiments provide a unique means to investigate microstructure development with negligible convection and to test the predictions of theoretical models in a purely diffusive growth regime. While most of the theoretical research to date has focused on the growth of single crystals, most commercially relevant metallic alloys are polycrystalline. The main aim of this research is to bring a quantum leap in our ability to model realistically dendritic alloy structures in polycrystalline materials where interactions between well-developed dendrites from the same grain and from grains of different crystal orientations both need to be considered. Interactions at grain boundaries are especially important because of their influence on dendrite fragmentation and the formation of solidification defects. The main goal of this project is to develop state-of-the-art computational tools to model dendritic microstructures formation under diffusive growth conditions and to establish benchmark comparisons with experimental observations under microgravity conditions. Those comparisons have the dual purpose to obtain new fundamental insights into microstructure formation and to validate computational models that can ultimately be used in an industrial setting to guide the design of structural alloys and optimize solidification processing conditions. Computational tools include the phase-field method and a new dendritic needle network (DNN) model. The phase-field method resolves the details of the dendrite tip scale and is ideally suited to model cellular and dendritic structures in dilute alloys on a scale comparable to the primary spacing. In the framework of a collaboration between NASA and the French National Space Agency (CNES), phase-field simulations are being used to model directional solidification experiments in a transparent organic succinonitrile–camphor alloy under low gravity in the Device for the Study of Critical Liquids and Crystallization (DECLIC) on-board the International Space Station (ISS). The DNN model in turn is designed to model dendritic microstructure formation on a grain scale that is orders of magnitude larger than the dendrite tip scale. Unlike previous models of grain structure evolution, the DNN approach models explicitly the growth kinetics of all branches of the hierarchical dendritic network (including primary, secondary, and tertiary branches) that interact diffusively via the solutal field. Hence, this method is ideally suited to investigate complex mechanisms of dendritic microstructure formation including the competitive growth of misoriented columnar grains and the columnar-to-equiaxed transition. This transition is of practical relevance for the casting industry. It is being investigated experimentally aboard the ISS in collaboration with our European partners in the framework of the Columnar-to-Equiaxed Transition in SOLidification Processing (CETSOL) project. Simulation results are critically important to interpret the results of those experiments in both metallic and transparent organic alloys and to guide future experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to lightweight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. This research is aimed at developing computational tools to predict reliably the formation of these dendritic microstructures on experimentally relevant length and time scales so as to accelerate and optimize the design of new technological alloys that can directly benefit US industries.
Task Progress:	During the current year, the following progress was achieved: CETSOL project We obtained a better understanding of the CET observed in experimental samples by applying the recently developed 3D extension of the DNN model. The DNN simulations used as input Batch 1 and Batch 2a temperature data to simulate the CET in a thin 3D geometry. The results were compared quantitatively to the measurements of the CET in refined Al-7 wt.% Si alloys solidified in microgravity under different thermal conditions. The DNN modeling predictions for the positions and spatial extent of the CET were found to be in reasonably good quantitative agreement with experimental measurements in two Batch 1 flight samples (FM1 and FM5) that exhibit sharp or progressive CET depending on the thermal conditions. Moreover, we extended our studies to bulk samples where the thickness of the sample is comparable with its width. The results were very close to the experimental values in terms of the starting and ending point of the CET. DECLIC project We performed three-dimensional phase-field simulations using a different description of the thermal field. In the simulations using the standard frozen temperature approximation (FTA) that is linear temperature profile within an adiabatic zone, the time of the morphological instability of the planar front is incorrectly predicted to be much shorter than that in the experiment. On the other hand, the instability time is well predicted when using the simulation with the thermal drift approximation (TDA) and the improved temperature field calculation (TFC), which includes latent heat rejection at the interface and thermal diffusion within an adiabatic zone. Additionally, we studied the selected primary spacings. We developed an analytical model to predict the initial spacing after the morphological instability. The model prediction shows a good agreement with the simulations and the experiments. For the final spacing at a steady state, the experimental measurements are close to the simulation with the TFC rather than the FTA. Those simulations reveal that the accurate calculation of the thermal field is essential to predict the time of the morphological instability and the final spacing selection, and thus improve our ability to predict this instability using accurate numerical models. We pursued our theoretical investigation of the dynamical selection of the primary spacing during the growth of misoriented crystals by our recently discovered source-sink mechanism. We further developed and tested our analytical model to predict the propagation of the front separating the region of large spacing near the divergent sample boundary, where new cells are created by tertiary branching (source), from the region of small spacing near the convergent boundary, where cell elimination occurs (sink). This model, which is based on a nonlinear advection equation, predicts that this front moves at about one third of the lateral drift velocity that is geometrically imposed by the crystal misorientation. This prediction is an excellent quantitative agreement with phase-field simulations and thin-sample terrestrial experiments. We investigated the solidification front dynamics near the convergent grain boundary of a 3D cellular array. In the DECLIC/DSI (Directional Solidification Insert) experiments, cellular arrays drift laterally due to their crystalline orientations. When grains, consisting of cells moving in a same direction, drift towards each other, one or both grains

	can penetrate into the other nearby grain, which leads to the instability of the grain boundary morphology. Moreover, experimental observations reveal that, during the penetration process, the leader cell of a penetrating group can detach itself from its own group and survive for a long time as a misoriented solitary cell within the other nearby grain. The phase-field (PF) simulations can capture those two dynamics near the convergent boundary. We investigated the spacing distribution patterns near the grain boundary. Experimental measurements show that the primary spacing increases near the divergent grain boundary while the spacing variation near the convergent boundary is difficult to notice due to the optical resolution. For this study, we developed the computational method to use multiple Graphics Processing Units (multi-GPUs), which allows us to perform simulations with a larger domain comparable to the experiments. In the preliminary simulations, the spacing near the divergent grain boundary is larger than the other region as observed in the experiments. Near the convergent grain boundary, however, the spacing decreases due to the elimination of the cells. We studied a dendrite shape and its morphological evolution due to oscillatory forcing of the pulling velocity. Preliminary simulations and DSI-R (Reflight) experiments reveal that an oscillation of a pulling velocity can accelerate the emergence of sidebranches. Specifically, PF simulations reveal that thermal effects suppress the growth of secondary branches, which explains the suppressed sidebranching amplitude in DSI-R experiments. On the other hand, we can identify the influence of oscillatory forcing in thin-sample experiments on Earth. Additionally, PF simulations show that oscillation period is linked to the sidebranching wavelength and the amplification of a tip radius oscillation. [EDITOR'S NOTE (February 2019): Project continues as "Computational Studies of Microstructure Formation during Alloy Solidification in Microgravity," grant 80NSSC19K0135, Principal Investigator Alain Karma, PhD]
Bibliography Type:	Description: (Last Updated: 05/10/2026)
Articles in Peer-reviewed Journals	Song Y, Tourret D, Mota FL, Pereda J, Billia B, Bergeon N, Trivedit R, Karma A. "Thermal-field effects on interface dynamics and microstructure selection during alloy directional solidification." Acta Materialia. 2018 May;150:139-52. https://doi.org/10.1016/j.actamat.2018.03.012 , May-2018
Articles in Peer-reviewed Journals	Pineau A, Guillemot G, Tourret D, Karma A, Gandin Ch-A. "Growth competition between columnar dendritic grains – cellular automaton versus phase field modeling." Acta Materialia. 2018 Aug;155:286-301. https://doi.org/10.1016/j.actamat.2018.05.032 , Aug-2018
Articles in Peer-reviewed Journals	Song Y, Akamatsu S, Bottin-Rousseau S, Karma A. "Propagative selection of tilted array patterns in directional solidification." Physical Review Materials. 2018 May;2:053403. https://doi.org/10.1103/PhysRevMaterials.2.053403 , May-2018
Articles in Peer-reviewed Journals	Song Y, Mota FL, Tourret D, Ji K, Billia B, Trivedi R, Bergeon N, Karma A. "Cell invasion during competitive growth of polycrystalline solidification patterns." Nat Commun. 2023 Apr 19;14(1):2244. https://doi.org/10.1038/s41467-023-37458-0 ; PMID: 37076477; PMCID: PMC10115863 , Apr-2023